

Assignment III - Discrete Math(H)

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Q.1

Q.1 What are the prime factorizations of

(1) 6560

(2) 12!

Sol:

1.
$$6560 = 2^5 \times 5 \times 41$$

2. Implementing the Legendre's Formula:

For base 2: $\lfloor 12/2 \rfloor + \lfloor 12/4 \rfloor + \lfloor 12/8 \rfloor = 10.$

For base 3: $\lfloor 12/3 \rfloor + \lfloor 12/9 \rfloor = 5.$

For base 5: $\lfloor 12/5 \rfloor = 2.$

For base 7: $\lfloor 12/7 \rfloor = 1.$

For base 11: $\lfloor 12/11 \rfloor = 1.$

Therefore, we have:

$$12! = 2^{10} \times 3^5 \times 5^2 \times 7 \times 11$$

Q.2

Q.2 Convert the decimal expansion of each of these integers to a binary expansion.

(a) 321 (b) 1023 (c) 100632

Sol:

1. We have $321 = 256 + 64 + 1 = 2^8 + 2^6 + 2^0 = (101000001)_2$.
2. We have $1023 = 1024 - 1 = 2^{10} - 2^0 = (1111111111)_2$.
3. With the regular method, we can calculate that $100632 = (11000100100011000)_2$.

Q.3

Q.3 For two integers a, b , prove that if $\gcd(a, b) = 1$, then

$$\gcd(b + a, b - a) \leq 2.$$

PF:

Let $d = \gcd(b + a, b - a) = \gcd(b + a, b - a + (b + a)) = \gcd(b + a, 2b)$, then we have $d \mid 2b$.

Let $d = \gcd(b + a, b - a) = \gcd(b + a - (b - a), b - a) = \gcd(2a, b - a)$, then we have $d \mid 2a$.

For $\gcd(a, b) = 1$, $\gcd(2a, 2b) = 2 \gcd(a, b) = 2$.

Therefore, $d \leq 2$, i.e., $\gcd(b + a, b - a) \leq 2$.

Q.E.D..

Q.4

Q.4

- (1) Give the prime factorization of 312.
- (2) Use Euclidean algorithm to find $\gcd(312, 97)$.
- (3) Find integers s and t such that $\gcd(312, 97) = 312s + 97t$.
- (4) Solve the modular equation

$$312x \equiv 3 \pmod{97}.$$

Sol:

1. We have $312 = 2^3 \times 3 \times 13$.

2. We have:

$$312 = 97 \times 3 + 21.$$

$$97 = 21 \times 4 + 13.$$

$$21 = 13 \times 1 + 8.$$

$$13 = 8 \times 1 + 5.$$

$$8 = 5 \times 1 + 3.$$

$$5 = 3 \times 1 + 2.$$

$$3 = 2 \times 1 + 1.$$

$$2 = 1 \times 2 + 0.$$

Therefore, $\gcd(312, 97) = 1$.

3. Implement the back-substitute:

$$1 = 3 - 2 \times 1.$$

$$1 = 3 - (5 - 3).$$

$$1 = 2 \times (8 - 5) - 5.$$

$$1 = 2 \times 8 - 3 \times (13 - 8).$$

$$1 = 5 \times (21 - 13) - 3 \times 13.$$

$$1 = 5 \times 21 - 8 \times (97 - 21 \times 4).$$

$$1 = 37 \times (312 - 97 \times 3) - 8 \times 97.$$

$$1 = 37 \times 312 - 119 \times 97.$$

Therefore, $s = 37, t = -119$.

4. For $312x \equiv 3 \pmod{97}$, and $312 \equiv 21 \pmod{97}$, we have: $21x \equiv 3 \pmod{97}$.

For (3), we have: $1 = 37 \times 21 - 8 \times 97$, then $37 \times 21 \equiv 1 \pmod{97}$, thus the inverse of 21 is 37, then $x \equiv 3 \times 37 \equiv 14 \pmod{97}$.

Therefore, $x \equiv 14 \pmod{97}$.

Q.5

Q.5 Suppose that p, q and r are distinct primes. Show that there exist integers a, b and c , such that

$$a(pq) + b(qr) + c(rp) = 1.$$

PF:

We have $\gcd(pq, qr) = q$ and $\gcd(q, rp) = 1$, for Bezout, we have: $s(pq) + t(qr) = q, uq + v(rp) = 1$.

Substitute the q of the second equation via the first equation: $us(pq) + ut(qr) + v(rp) = 1$.

W.l.o.g., let $a = us, b = ut, c = v$, it's obvious to prove.

Q.E.D..

Q.6

Q.6 Prove that if a and m are positive integers such that $\gcd(a, m) \neq 1$ then a does *not* have an inverse modulo m .

PF:

Let $d = \gcd(a, m) \neq 1$, we have $d > 1$ and $d \mid a$ and $d \mid m$.

Assuming a had an inverse, i.e., $\exists x$ s.t. $ax \equiv 1 \pmod{m}$, i.e., $m \mid (ax - 1)$.

For $d \mid a$ and $d \mid m$, then $d \mid ax$, and $d \mid (ax - 1)$, then $d \mid (ax - 1) - ax = -1$, i.e., $d \mid 1$, leading to a contradiction.

Thus, a has no inverse mod m .

Q.E.D..

Q.7

Q.7 Prove that there are infinitely many primes of the form $6k + 5$.

PF:

Assume that there are finitely many primes of the form $6k + 5$, denote as $p_1, p_2, \dots, p_n$, let $P = p_1 \cdots p_n$ and let $N = 6P - 1$.

Then $N \equiv 5 \pmod{6}$ and for each i we have $N \equiv -1 \pmod{p_i}$, thus $p_i \nmid N$.

Let q be a prime divisor of N . Since $q > 3$, obviously there has to be $q \equiv 1$ or $q \equiv 5 \pmod{6}$.

If all prime factors s.t. $q \equiv 1 \pmod{6}$, their product would s.t. $\prod q \equiv 1 \pmod{6}$, leading to a contradiction to $N \equiv 5 \pmod{6}$.

Thus some factor satisfies $q \equiv 5 \pmod{6}$. Since $q \nmid P$, which is not in the prime list, producing a new $6k + 5$ prime, leading to a contradiction.

Thus, there are infinitely many primes of the form $6k + 5$.

Q.E.D..

Q.8

Q.8 Let a and b be positive integers. Show that $\gcd(a, b) + \text{lcm}(a, b) = a + b$ if and only if a divides b , or b divides a .

PF:

Let $d = \gcd(a, b)$ and let $a = dx$, $b = dy$ with $\gcd(x, y) = 1$.

Then $\text{lcm}(a, b) = \frac{ab}{\gcd(a, b)} = dxy$.

$$\begin{aligned}\gcd(a, b) + \text{lcm}(a, b) &= a + b \\ \iff d + dxy &= dx + dy \\ \iff 1 + xy &= x + y \\ \iff (x - 1)(y - 1) &= 0\end{aligned}$$

And $x = 1$ or $y = 1$ is obviously represent $a \mid b$ or $b \mid a$.

Q.E.D..

Q.9

Q.9

(1) Show that there is no integer solution x to the equation

$$x^2 \equiv 31 \pmod{36}.$$

(2) Find the integer solutions x to the system of equations

$$\begin{cases} x^2 \equiv 10 \pmod{31}, \\ x^2 \equiv 30 \pmod{37}. \end{cases}$$

1. PF:

Since 36 is divisible by 4, any solution would also satisfy: $x^2 \equiv 31 \equiv 3 \pmod{4}$.

But squares modulo 4 are only 0 or 1, leading to a contradiction.

Thus there is no integer solution.

Q.E.D..

2. Sol:

For the first equation, the roots mod 31 are $x \equiv 14, 17$.

For the second equation, the roots mod 37 are $x \equiv 17, 20$.

Combine them via CRT:

Let $M = 31 \cdot 37 = 1147$. Inverses:

$$37 \equiv 6 \pmod{31}, 37^{-1} \equiv 26 \pmod{31}.$$

$$31 \equiv -6 \pmod{37}, 31^{-1} \equiv 6 \pmod{37}.$$

Thus $c_1 = 37 \times 26$, $c_2 = 31 \times 6$.

- $(14, 17) \Rightarrow x \equiv 572 \pmod{1147}$,
- $(17, 17) \Rightarrow x \equiv 17 \pmod{1147}$,
- $(14, 20) \Rightarrow x \equiv 1130 \pmod{1147}$,
- $(17, 20) \Rightarrow x \equiv 575 \pmod{1147}$.

Therefore, $x \equiv 17, 572, 575, 1130 \pmod{1147}$.

Q.10

Q.10 Compute the following without calculator. You may find Fermat's little theorem useful for some of these.

(1) The last decimal digit of 3^{1000}

(2) $3^{1000} \pmod{31}$

(3) $3/16$ in $\mathbb{Z}_{31}$

Sol:

1. The cycles 3, 9, 7, 1 with $T = 4$. Since $1000 \equiv 0 \pmod{4}$, the last digit is 1.
2. By Fermat's little theorem, $3^{30} \equiv 1 \pmod{31}$, and $1000 \equiv 10 \pmod{30}$, so $3^{1000} \equiv 3^{10} \pmod{31}$. And $3^{10} \equiv 25 \pmod{31}$.

Thus $3^{1000} \equiv 25 \pmod{31}$.

3. Since $16 \cdot 2 \equiv 1 \pmod{31}$, we have $16^{-1} \equiv 2 \pmod{31}$, thus

$$3/16 \equiv 3 \cdot 16^{-1} \equiv 3 \cdot 2 \equiv 6 \pmod{31}.$$

Q.11

Q.11 Show that if a and m are relatively prime positive integers, then the inverse of a modulo m is unique modulo m .

PF:

Assume $\gcd(a, m) = 1$ and x, y are both inverses of a modulo m .

We have:

$$a(x - y) \equiv 0 \pmod{m}.$$

Hence $m \mid a(x - y)$. Since $\gcd(a, m) = 1$, we have $m \mid (x - y)$, i.e. $x \equiv y \pmod{m}$.

Thus the inverse of a modulo m is unique modulo m .

Q.E.D..

Q.12

Q.12 Let $m_1, m_2, \dots, m_n$ be pairwise relatively prime integers greater than or equal to 2. Show that if $a \equiv b \pmod{m_i}$ for $i = 1, 2, \dots, n$, then $a \equiv b \pmod{m}$, where $m = m_1 m_2 \cdots m_n$.

PF:

If $\gcd(u, v) = 1$ and $u \mid (a - b)$, $v \mid (a - b)$, then obviously $uv \mid (a - b)$.

Thus for every m_i , they all s.t. $m_i \mid (a - b)$, and all the m_i are relatively prime, thus their product m also satisfy $m \mid (a - b)$, i.e., $a \equiv b \pmod{m}$.

Q.E.D..

Q.13

Q.13 Find all solutions, if any, to the system of congruences $x \equiv 5 \pmod{6}$, $x \equiv 3 \pmod{10}$, and $x \equiv 8 \pmod{15}$.

Sol:

With the coprime-factor rule, we have:

$$\begin{aligned}x \equiv 5 \pmod{6} &\Rightarrow x \equiv 1 \pmod{2}, x \equiv 2 \pmod{3} \\x \equiv 3 \pmod{10} &\Rightarrow x \equiv 1 \pmod{2}, x \equiv 3 \pmod{5} \\x \equiv 8 \pmod{15} &\Rightarrow x \equiv 2 \pmod{3}, x \equiv 3 \pmod{5}\end{aligned}$$

Then Implement CRT:

$$M = 30.$$

$$M_1 = 15, M_2 = 10, M_3 = 6.$$

$$M_1^{-1} = 1, M_2^{-1} = 1, M_3^{-1} = 1.$$

Eventually, we have $x \equiv 23 \pmod{30}$.

Tips: Implementing exCRT can also solve this problem.

Q.14

Q.14 Recall how the *linear congruential method* works in generating pseudorandom numbers: Initially, four parameters are chosen, i.e., the modulus m , the multiplier a , the increment c , and the seed x_0 . Then a sequence of numbers $x_1, x_2, \dots, x_n, \dots$ are generated by the following congruence

$$x_{n+1} = (ax_n + c) \pmod{m}.$$

Suppose that we know the generated numbers are in the range $0, 1, \dots, 10$, which means the modulus $m = 11$. By observing three consecutive numbers 7, 4, 6, can you predict the next number? Explain your answer.

Sol:

From $x_2 \equiv ax_1 + c \pmod{11}$ and $x_3 \equiv ax_2 + c \pmod{11}$, subtract :

$$x_3 - x_2 \equiv a(x_2 - x_1) \pmod{11}$$

We have: $x_3 - x_2 = 2$, $x_2 - x_1 = -3 \equiv 8$, and $8^{-1} \equiv 7$.

Then,

$$a \equiv 2 \cdot 8^{-1} \equiv 2 \cdot 7 \equiv 14 \equiv 3 \pmod{11}$$

Thus,

$$c \equiv x_2 - ax_1 \equiv 4 - 3 \cdot 7 \equiv 4 - 21 \equiv -17 \equiv 5 \pmod{11}.$$

Then, $x_4 \equiv 3 \cdot 6 + 5 \equiv 23 \equiv 1 \pmod{11}$.

Q.15

Q.15

- (1) Use Fermat's little theorem to compute $5^{2003} \pmod{7}$, $5^{2003} \pmod{11}$, and $5^{2003} \pmod{13}$.
- (2) Use your results from part (a) and the Chinese remainder theorem to find $5^{2003} \pmod{1001}$. (Note that $1001 = 7 \cdot 11 \cdot 13$.)

Sol:

Q15

1.

Since $5^6 \equiv 1 \pmod{7}$:

$$5^{2003} \equiv 5^5 \equiv 3 \pmod{7}$$

Since $5^{10} \equiv 1 \pmod{11}$:

$$5^{2003} \equiv 5^3 \equiv 4 \pmod{11}$$

Since $5^{12} \equiv 1 \pmod{13}$:

$$5^{2003} \equiv 5^{11} \equiv 8 \pmod{13}$$

2. Implement CRT to modulus $1001 = 7 \cdot 11 \cdot 13$:

Let $M = 1001$, $M_1 = 143$, $M_2 = 91$, $M_3 = 77$.

Inverses

$$\begin{aligned} M_1 &\equiv 3 \pmod{7} \Rightarrow M_1^{-1} \equiv 5 \pmod{7} \\ M_2 &\equiv 3 \pmod{11} \Rightarrow M_2^{-1} \equiv 4 \pmod{11} \\ M_3 &\equiv -1 \pmod{13} \Rightarrow M_3^{-1} \equiv 12 \pmod{13} \end{aligned}$$

Set $c_1 = 143 \cdot 5 = 715$, $c_2 = 91 \cdot 4 = 364$, $c_3 = 77 \cdot 12 = 924$.

Then

$$x \equiv 3 \cdot 715 + 4 \cdot 364 + 8 \cdot 924 = 10993 \equiv 983 \pmod{1001}$$

Therefore, $5^{2003} \equiv 983 \pmod{1001}$.

Q.16

Q.16 Given an integer a , we say that a number n passes the “Fermat primality test (for base a)” if $a^{n-1} \equiv 1 \pmod{n}$.

- (1) For $a = 2$, does $n = 561$ pass the test?
- (2) Did the test give the correct answer in this case?

PF:

1. Factor $561 = 3 \cdot 11 \cdot 17$. Via Fermat’s little theorem:

$$\begin{aligned} 2^{560} &= (2^2)^{280} \equiv 1 \pmod{3} \\ 2^{560} &= (2^{10})^{56} \equiv 1 \pmod{11} \\ 2^{560} &= (2^{16})^{35} \equiv 1 \pmod{17} \end{aligned}$$

By CRT, $2^{560} \equiv 1 \pmod{561}$. Hence 561 passes the test.

2. No. For 561 is composite: $561 = 3 \cdot 11 \cdot 17$.

Q.17

Q.17 Show that we can easily factor n when we know that n is the product of two primes, p and q , and we know the value of $(p-1)(q-1)$.

PF:

We know $n = pq$ and $\varphi(n) = (p-1)(q-1) = n - (p+q) + 1$.

Hence:

$$s = p + q = n - \varphi(n) + 1.$$

Then p and q solve

$$X^2 - sX + n = 0.$$

$$D = s^2 - 4n = (p+q)^2 - 4pq = (p-q)^2.$$

Let $r = \sqrt{D}$:

$$p = \frac{s+r}{2} \quad \text{and} \quad q = \frac{s-r}{2}.$$

Q.E.D..

Q.18

Q.18 Consider the RSA encryption method. Let our public key be $(n, e) = (65, 7)$, and our private key be d .

- (1) What is the encryption $\hat{M}$ of a message $M = 8$?
- (2) To decrypt, what value d do we need to use?
- (3) Using d , run the RSA decryption method on $\hat{M}$.

1. Encryption of $M = 8$

$$\hat{M} \equiv M^e \pmod{n}, \quad \hat{M} \equiv 8^7 \pmod{65}$$

Since $8^2 \equiv 64 \equiv -1 \pmod{65}$

$$8^7 = 8(8^2)^3 \equiv 8(-1)^3 \equiv -8 \equiv 57 \pmod{65}$$

Thus $\hat{M} = 57$.

2. Decryption exponent

$$\varphi(65) = (5-1)(13-1) = 48, \quad 7d \equiv 1 \pmod{48}, \quad d = 7$$

3. Decrypt

$$\hat{M}^d \equiv 57^7 \pmod{65}, \quad 57 \equiv -8 \pmod{65}, \quad 8^2 \equiv -1 \pmod{65}$$

$$57^7 \equiv (-8)^7 \equiv -8^7 \equiv -57 \equiv 8 \pmod{65}$$

Result: $M = 8$.

Q.19

Q.19 Consider the RSA system. Let (e, d) be a key pair for the RSA. Define

$$\lambda(n) = \text{lcm}(p-1, q-1)$$

and compute $d' = e^{-1} \bmod \lambda(n)$. Will decryption using d' instead of d still work? (prove $C^{d'} \bmod n = M$)

Let $n = pq$ with primes p, q . Define $\lambda(n) = \text{lcm}(p-1, q-1)$ and choose d' with $ed' \equiv 1 \pmod{\lambda(n)}$.

For any message M and ciphertext $C \equiv M^e \pmod{n}$ show $C^{d'} \equiv M \pmod{n}$.

Let $ed' = 1 + k\lambda(n)$. Prove modulo p and modulo q .

- If $p \mid M$ then $C \equiv 0 \pmod{p}$ hence $C^{d'} \equiv 0 \equiv M \pmod{p}$.
- If $\gcd(M, p) = 1$ then $p-1 \mid \lambda(n)$ thus:

$$\begin{aligned} M^{\lambda(n)} &\equiv 1 \pmod{p} \\ M^{ed'} &= M^{1+k\lambda(n)} \equiv M \cdot (M^{\lambda(n)})^k \equiv M \pmod{p}. \end{aligned}$$

The same holds modulo q . Therefore $C^{d'} \equiv M \pmod{p}$ and $C^{d'} \equiv M \pmod{q}$.

Via CRT we conclude $C^{d'} \equiv M \pmod{n}$. Thus decryption using d' works.